Multivariate Calculus Lecture Notes
Section 14.1 – Functions of Two or More Variables

I. Domains and ranges
   a. Definitions
      i. Let $D \subseteq \mathbb{R}^2$.
      ii. Let f be a function that assigns to each ordered pair (x, y) in D exactly one number $z = f(x, y)$.
      iii. Then D is the *domain* of f.
      iv. And the set of all values taken on by z is the *image* of f. (Your text uses *range*, but *range* can be a bit ambiguous, depending on the context.)
      v. Unless otherwise specified, the domain of f is assumed to be all pairs (x, y) which produce some real number for z.
      vi. These ideas can apply to any subset of $\mathbb{R}^n$ for any n.
   b. Examples
      i. Find the domain of
$$f(x, y) = \frac{\sqrt{x + y + 1}}{x - 1}.$$
         1. $x + y + 1 \geq 0$
         2. $x - 1 \neq 0$
         3. Algebraically, $D = \{(x, y) : x + y + 1 \geq 0, x \neq 1\}$.
         4. Graphically:

      ii. Find the domain of $f(x, y) = x \ln(y^2 - x)$.
         1. $y^2 - x > 0$
         2. $D = \{(x, y) : y^2 - x > 0\}$
         3. Graphically:

II. Graphs
   a. $f(x, y) = 6 - 3x - 2y$
      i. This is $z = 6 - 3x - 2y$, which we know to be a plane.
      ii. Graphically:

      iii. Domain: $\mathbb{R}^2$
      iv. Range: $\mathbb{R}$
   b. $g(x, y) = \sqrt{9 - x^2 - y^2}$
      i. This is $z = \sqrt{9 - x^2 - y^2}$.

ii. This is the top half of a sphere of radius 3, as squaring both sides plus a little algebra produces $x^2 + y^2 + z^2 = 9$.
iii. Graphically:

iv. Domain: The disk of radius 3 centered at the origin.
v. Range: $[0, 3]$

c. Graph the following using Mathematica.
i. $f(x, y) = \sin x$
ii. $f(x, y) = \sin x \sin y$
iii. $f(x, y) = \sin x \sin y/(xy)$
iv. $f(x, y) = \sin x + \sin y$
v. $f(x, y) = e^{-x^2-y^2}$
vi. $f(x, y) = xye^{-x^2-y^2}$
vii. $f(x, y) = \sin(\sqrt{x^2 + y^2})$
viii. $f(x, y) = -1/\sqrt{x^2 + y^2}$

III. Contour diagram basics
a. Another way to represent multivariable functions is with contour diagrams, or level curves.
b. The idea is that every point with a particular value lies on the same curve.
c. The most well-known type of contour diagram is probably the topographic map.
d. Show topographic map of Fuller Mountain (east of Seattle) and discuss some of its basic features.
i. Where is the land the steepest?
ii. Where is the land the flattest?
iii. Where is a ridge line?
iv. Where is a saddle?

IV. Sketching contour diagrams
a. Draw the contour diagram for $f(x, y) = 4x^2 + y^2$.
i. Ask students what this will look like.
ii. Examine the function for different values of $z = f(x, y)$.
1. $z = 0$: the point $(0, 0)$
2. $z = 1$: an ellipse
3. $z = 2$: an larger ellipse
4. $z = 3$: an even larger ellipse
5. $z = -1$: this cannot exist in real three-space
iii. Graphically:

b. Draw the contour diagram for $f(x, y) = 6 - 3x - 2y$.
i. Ask students what this will look like.

ii. $z = 0$: $3x + 2y = 6$
iii. $z = 6$: $3x + 2y = 0$
iv. $z = 12$: $3x + 2y = -6$
v. $z = -6 : 3x + 2y = 12$
vi. Graphically:

vii. If you're on the level curve, which direction is the steepest?
    1. Answer: perpendicular to the level curve.
    2. The same is true for the other types of contour diagrams.
    3. This will come up again later.
viii. If you're on the level curve, which direction would you go in order to stay at the same height?
    1. Answer: parallel to the level curve.
    2. This will also come up again later.

c. Use the contour diagram to graph $f(x, y) = x + y^2$ in three dimensions.
    i. Contour diagram:

    ii. Three dimensions:

    iii. It's an ever-increasing parabola, kind of like a slide. (Illustrate with a piece of paper.)

d. It's important to pick values of z that have the same distance between them to get a good feel for where the graph is and is not steep.

V. Interpreting contour diagrams
a. What could the surfaces represented by these contour diagrams look like?
    i. Plane

ii. Parabolic cylinder

b. Draw contour diagram of $z = y - \sin x$ using Mathematica. (Larger values are shown lighter, and smaller values are shown darker. See Figure 1.)

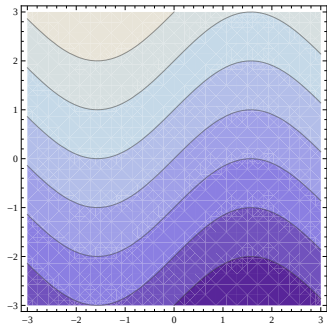


FIGURE 1. $z = y - \sin x$

i. If you walk along the line $x = \pi/2$, are you on a ridge or a valley? (a valley)
ii. Where might you be on a ridge? ($x = -\pi/2$)

VI. Level surfaces
a. Even though we can't graph functions of three variables we can illustrate them through the use of level surfaces.
b. Example
i. Find the level surfaces of the function $f(x, y, z) = x^2 + y^2 + z^2$.
ii. The level surfaces are concentric spheres.

Homework: Section 14.1 #7, 11, 19, 20, 24, 30, 33, 38, 46.